# ANALYTICAL CUMULATIVE EXAM
October 28, 1995

## Electrochemistry

Chose *one* of the following electrochemical techniques and answer the questions below.

cyclic voltammetry
linear voltammetry
double step chronoamperometry
chronocoulometry
pulse voltammetry
time-based constant potential electrolysis
rotating disk or rotating ring-disk hydrodynamic methods

1. (15 pts.) Describe the operational features of the technique (what parameters are controlled, what is the applied function of current or potential, and what parameters are measured).

2. (25 pts.) Describe the features of the monitored response and plots of data that can be obtained from the technique when working with an electrolyte solution with a stable, kinetically reversible (where n = 1), non-adsorbing redox couple (such as $Ru(NH_3)_6^{2+/3+}$) for which the standard reduction potential is E°. What information do the response and plotted representations of the data provide? How do you extract the information from the plots? To support your discussion, include examples of plots, raw data, and equations.

3. (40 pts. total, 5 pts. each) What would be the expected response if the following changes were made?

(a) the double layer capacitance is increased
(b) the resistance of the solution is increased
(c) the concentration of the redox couple is increased
(d) the diffusion coefficient of both oxidized and reduced species increase by the same amount
(e) the temperature is increased
(f) the number of electrons per mole of redox species (n) is increased
(g) the time scale of the electrochemical experiment is increased.
(h) the electrochemical product adsorbs strongly, but the reactant does not.

4. (10 pts.) Compare the responses you would expect for the case when (1) the thickness of the diffusion layer is much less than the smallest dimension of the electrode, and for the case when (2) the thickness of the diffusion layer is much greater than the smallest dimension of the electrode? Use equations to support your discussion, if necessary. (Hint: This question compares macroelectrode behavior with microelectrode behavior.)

5. (10 pts.) In general, what are the advantages and disadvantages of the technique over other electrochemical techiques?